

In this note, I write down some facts related to Barron Class and Weighted Variation Spaces. These are useful function spaces for the study of approximation theory for neural networks. The ideas are from DeVore et. al (2025).

1 Barron Class

In 1993, Andrew Barron introduced the Barron class $\mathcal{B}^s$, where s is a smoothness index on Fourier moments. On $\mathbb{R}^d$, the set $\mathcal{B}^s(\mathbb{R}^d)$ consists of all functions f defined on $\mathbb{R}^d$ whose Fourier transform $\hat{f}$ satisfies

$$\|f\|_{\mathcal{B}^s} := \int_{\mathbb{R}^d} (1 + |\omega|)^s |\hat{f}(\omega)| d\omega < +\infty.^1$$

The point of the Barron class, broadly speaking, is that on a bounded domain $\Omega \subset \mathbb{R}^d$, any function f on Ω in $\mathcal{B}^1$ can be approximated in the $L_2(\Omega)$ norm by a single hidden layer sigmoid neural network with n neurons to the accuracy of

$$C_\Omega \|f\|_{\mathcal{B}^1} n^{-1/2}.$$

This approximation rate does not deteriorate with increasing d , but the definition of Barron space does get more demanding as d increases (because of the d -dimensional integral). In this sense, Barron classes do not eliminate the dimension dependence, but relocate it out of the exponent in the rate into the model class assumption.

2 Weighted Variation Spaces

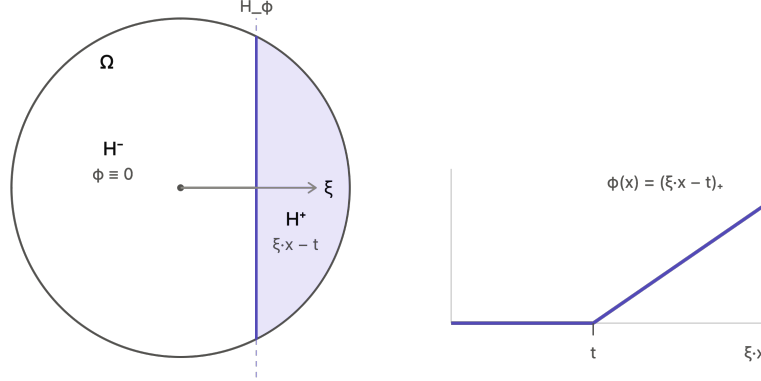
I start with a few definitions.

1. The ReLU activation function is $\sigma(t) = \max(0, t)$.
2. $\Omega \subset \mathbb{R}^d$ is a convex, bounded domain.

¹ As a simple example, consider the function $g(x) = e^{-x^2/2}$, which has the Fourier transform $\hat{g}(\omega) = \int_{-\infty}^{\infty} g(x) e^{-2\pi i \omega x} dx$. The Barron class norm with $s = 1$ is

$$\|g\|_{\mathcal{B}^1} = \int_{-\infty}^{\infty} (1 + |\omega|) \left| \int_{-\infty}^{\infty} e^{-x^2/2} e^{-2\pi i \omega x} dx \right| d\omega = 1 + \frac{1}{\pi\sqrt{2\pi}} < \infty.$$

3. The ReLU **atom** is defined by $\phi(x; \xi, t) := \sigma(\xi x - t)$ where $\xi \in \mathbb{R}^d, \|\xi\| = 1, t \in \mathbb{R}$.
4. The **hyperplane cut** is $H_\phi := x \in \Omega : \xi x = t$. This splits Ω into H_ϕ^- where $\phi = 0$ and H_ϕ^+ , where $\phi = \xi x - t$ affine function.



5. The active parameters are defined by

$$Z(\Omega) := \{(\xi, t) : \xi \in \mathbb{R}^d, \|\xi\| = 1, t \in \mathbb{R} \text{ such that } H_{\phi(\cdot; \xi, t)} \cap \Omega \neq \emptyset\}.$$

Note that the closure $\bar{Z}(\Omega)$ is compact.

6. The **dictionary** is $\mathcal{D}(\Omega) := \{\phi(\cdot; \xi, t) : (\xi, t) \in \bar{Z}(\Omega)\}$. Since ϕ is continuous and $\bar{Z}(\Omega)$ is compact, $\mathcal{D}(\Omega)$ is a compact subset of $L_2(\Omega)$. As an example, with $\Omega = B^d$ the d -dimensional Euclidean ball, the atom ϕ has $L_2(\Omega)$ -norm satisfying $\|\phi\|_{L_2(\Omega)} \approx (1 - t)^{\frac{3}{2} + \frac{d-1}{4}}$ for $-1 \leq t \leq 1$.

EXAMPLE DERIVATION TODO.

7. The set of functions that give a n -term approximation (or equivalently, the width- n single layer ReLU network) is given by

$$\Sigma_n := \Sigma_n(\mathcal{D}) = \left\{ \sum_{j=1}^n a_j \phi_j : \phi_j \in \mathcal{D}, a_j \in \mathbb{R} \right\}$$

8. For a particular $S \in \Sigma_n$, its **variation** is given by

$$V(S) := \inf \left\{ \sum_{j=1}^n |a_j| : S = \sum_{j=1}^n a_j \phi_j \right\}.$$

The **variation space** for the dictionary $\mathcal{D}$ is defined by

$$\mathcal{V}(\Omega, \mathcal{D}) := \left\{ f \in L_2(\Omega) : \exists S_n \in \Sigma_n \text{ s.t. } \|f - S_n\|_{L_2(\Omega)} \rightarrow 0 \text{ and } V(S_n) \leq M \ \forall n \right\}$$

where M is a constant. The smallest M for which this holds is defined by $\|f\|_{\mathcal{V}(\Omega)}$. The variation space is a Banach space, and a fundamental property is that

$$\|f\|_{\mathcal{V}(\Omega)} \leq C_\Omega \|f\|_{\mathcal{B}^2(\Omega)}$$

so that the variation space is larger than the Barron class. Furthermore, $\mathcal{V}(\Omega)$ is equal to Radon-domain BV spaces (i.e. functions with a Radon transform in a certain smoothness space) up to norm equivalence.

9. The **weight function** is $w(\xi, t)$ with $(\xi, t) \in \bar{Z}(\Omega)$, and it is a non-negative continuous function on $\bar{Z}(\Omega)$. Correspondingly the normalized atom is $\tilde{\phi}(\cdot; \xi, t) := \frac{\phi(\cdot; \xi, t)}{w(\xi, t)}$ and it is zero whenever $w(\xi, t) = 0$. We say that w is **admissible** for Ω if $(\xi, t) \mapsto \tilde{\phi}(\cdot; \xi, t)$ is continuous as a mapping $\bar{Z}(\Omega) \rightarrow L_2(\Omega)$. By compactness, the norm

$$\|\tilde{\phi}(\cdot; \xi, t)\|_{L_2(\Omega)} \leq C_w < \infty$$

for some constant C_w and so the new normalized dictionary

$$\mathcal{D}_w := \left\{ \tilde{\phi}(\cdot; \xi, t) : (\xi, t) \in \bar{Z}(\Omega) \right\}$$

is a compact subset of $L_2(\Omega)$.

We are now ready to state the main definition, the weighted variation space.

Definition 2.1 (Weighted Variation Space). *The weighted variation space $\mathcal{V}_w := \mathcal{V}_w(\Omega) := V(\mathcal{D}_w)$ is the variation space of the normalized dictionary $\mathcal{D}_w$.*

Next, for a function f in $L_p(\Omega)$ with $1 \leq p \leq \infty$, define the error of the best

approximation of f by a width- n single layer ReLU network as

$$E_n(f)_p := E_n(f)_{L_p(\Omega)} := \inf_{S \in \Sigma_n} \|f - S\|_{L_p(\Omega)}.$$

DeVore (2025) proves the following theorem, an approximation result on the domain $\Omega = B^d$, $d > 2$.

Theorem 1. *If $f \in \mathcal{V}_\omega = \mathcal{V}_\omega(\Omega)$ then*

$$E_n(f) \leq C \|f\|_{\mathcal{V}_\omega} n^{-\frac{1}{2} - \frac{3}{2d}}, \quad n \geq 1.$$

where C only depends on d .